

A project proposal for partial fulfilment of the course unit CSC3232-
Group Project for the degree of Bachelor of Science in Applied Mathematics
and Computing

Student Project and Research Archive Management System

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Declaration

We hereby declare that the project proposal submitted for evaluation of the course module CSC3232 leading to the award of a Bachelor of Science in Applied Mathematics and Computing is entirely our own work, and the contents taken from the work of others have been cited and acknowledged within the text. This proposal has not been submitted for any degree at this University or any other institution.

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I recommend that the project be carried out by the students under my supervision,

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Supervisor,

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Date

1. Introduction

1.1 Background

Student projects and research activities play a crucial role in university education, as they help students develop practical knowledge, research skills, and problem-solving abilities. Every academic year, numerous student groups complete projects and research work that generate valuable information and data. However, without a proper centralized system, these project records and research materials are often scattered and difficult to manage, making it challenging for departments and future students to access previous work.

In many university departments, student projects and research activities are often managed without a centralized digital system. Project records and research data are typically stored using separate files, paper documents, or informal digital storage methods. While these approaches may be manageable on a small scale, they become inefficient when handling multiple batches, departments, and research works over time.

Some common challenges include:

- Difficulty in tracking and maintaining past student project records.
- Research data stored in multiple locations, making retrieval difficult.
- Lack of a centralized archive for organizing academic projects.
- Time-consuming manual record management for the department.
- Limited access for students to review previous research and projects.

Particularly in the Faculty of Applied Science at the University of Vavuniya, there are several project-related modules each year, including group projects, software development projects, and research projects across various degree programs such as Applied Mathematics and Computing, Computer Science, Information Technology, and Environmental Science. Currently, there is no proper mechanism to systematically store, manage, or reuse these project outputs and associated data.

In addition to project reports, many research and software projects involve various types of research data and development resources, such as datasets (e.g., medical or biological data

collected with ethical approvals), documents, videos, audio files, and source codes used to implement the projects. These resources are valuable for understanding how the research was conducted and how the project was developed. However, without a proper system, these materials are often difficult to access and may not be preserved for future use. To address this, the system will incorporate a separate special storage device (e.g., a dedicated file server or NAS) for archiving sensitive or large datasets, managed by authorized department personnel to ensure security and compliance.

Therefore, it is essential to develop a comprehensive system to efficiently manage these projects and research resources and make them accessible for future students and academic purposes. The proposed system will organize project information based on department, academic year, and project group, while also storing important details such as project titles, group members, registration numbers, and research data used in the project.

These issues highlight the need for a digital solution that can centralize student project and research management, improve organization, and make academic information easily accessible to all users in the Faculty of Applied Science, University of Vavuniya.

This project proposes the development of a Student Project and Research Archive Management System (SPRAMS), a web-based platform designed to store, organize, and manage student projects and research data in a structured and secure manner. The system will allow administrators to maintain clear records of projects and research, while lecturers and students can easily view archived information. By creating a centralized digital repository with integrated special storage for archives, the system supports knowledge sharing, helps future students understand past research, access project resources such as datasets and source codes, and improves the efficiency of departmental data management.

1.2 Aim and Objective

The aim of this project is to design and develop a web-based Student Project and Research Archive Management System for the Faculty of Applied Science, University of Vavuniya, to store, organize, and manage student projects, research information, and research data in a centralized digital platform. The system will enable administrators to maintain structured academic records while allowing lecturers and students to easily access previous projects, research resources, and implementation details for learning and reference purposes.

The specific objectives of the system are:

1. Student Project Management – To develop a system for storing and organizing student project information based on department, academic year, group number, and project details.
2. Research Archive Management – To create a centralized repository for storing and managing research works and their related information.
3. Research Data Storage – To allow administrators to store research-related data such as documents, audio files, video files, and other datasets (e.g., medical or biological data) used during research, utilizing a separate special storage device for secure archiving.
4. User Role Management – To implement role-based access for Admin, Lecturer, and Student users to ensure proper control and management of the system.
5. Archive Browsing and Search – To enable users to easily browse archived projects and research based on department, academic year, and group information.
6. Lecturer Feedback Support – To allow lecturers to review project details and provide feedback for academic improvement.
7. Notification System – To notify users when new projects or research records are added to the archive.

1.3 Benefits of the System

The proposed Student Project and Research Archive Management System will provide several benefits to the Faculty of Applied Science, departments, lecturers, and students:

1. Centralized Project and Research Management – All student projects, research records, and related data can be stored and managed in a single digital platform.
2. Easy Access to Previous Work – Lecturers and students can easily view archived projects, research information, and implementation details for academic reference.
3. Better Organization of Academic Records – Projects and research are systematically arranged by department, academic year, and group information, ensuring clear and structured documentation.
4. Access to Project Implementation Resources – Students and lecturers can access GitHub repository links to understand project implementation and source code development.
5. Research Data Availability – Research-related materials such as documents, audio files, video files, and other datasets can be stored and accessed for learning and reference, with secure management by authorized personnel.

6. Improved Knowledge Sharing – Future students can study previous projects and research to gain ideas, understand methodologies, and improve new project development.
7. Efficient Department Management – The system helps departments maintain accurate records and manage project information more efficiently.
8. Notification of New Projects – Users can receive notifications when new projects or research records are added, keeping them updated with the latest academic work.
9. Secure Digital Storage – Important academic documents such as project proposals, presentation slides, and research files are securely stored for long-term use, including on a dedicated archive storage device.
10. Project Idea Generation for Students – Students can view completed, ongoing, and unfinished projects from previous academic years. This helps them understand existing work, identify research gaps, and generate new ideas for developing improved or related projects in the future.

2. Review of the Existing Systems

(a) Development of a Final Year Students Project Repository System [1]

The paper proposes an online repository system for final year projects, enabling students to easily access and share academic resources, promoting knowledge sharing and supporting academic research.

(b) Implementing lean principles for digital archiving of student projects [2]

Lean principles can effectively transition higher education institutions from paper-based project documentation to a sustainable, paperless digital system, reducing annual printed project books.

(c) A Web-Based Academic Project Management and Evaluation System [3]

Thesis IT is a web-based platform that streamlines project submission and evaluation for students, reducing administrative burden and ensuring secure storage of academic work.

Some limitations of these existing systems in an academic project context are:

- They do not provide a dedicated structure for organizing student projects by department, academic year, and group.

- Research data is often stored without proper categorization, making it difficult to search and retrieve.
- They lack customized role-based access control for administrators, lecturers, and students.
- They do not integrate project and research management into a single centralized platform.
- Many systems focus only on file storage rather than long-term academic record management.

Therefore, while existing tools show the importance of digital storage systems, they do not fully meet the specific needs of managing student projects and research archives in a university department. The proposed system aims to provide a customized and organized solution designed specifically for academic project and research management.

3. Materials and Methods

3.1 Overview of the Student Project and Research Archive Management System

The Student Project and Research Archive Management System is designed to address the lack of a systematic mechanism for storing, managing, and reusing student project outputs and research data at the Faculty of Applied Science, University of Vavuniya. This web-based system will serve as a centralized repository for various project types across degree programs Applied Mathematics and Computing, Computer Science, Information Technology, and Environmental Science.

Key features include:

- **User Roles and Access Control:** Students can upload project reports, code, datasets (e.g., medical or biological data collected with ethical approvals), and other outputs. Faculty and authorized department personnel can review, approve, manage, and access archives. Admins handle system maintenance.

- **Data Repository:** A secure, searchable database for metadata (e.g., project titles, authors, abstracts, keywords) integrated with a separate special storage device (e.g., a dedicated file server, NAS, or cloud-based archive like AWS S3 with encryption) for storing sensitive or large datasets to ensure compliance with ethical standards and data privacy.

- **Functionalities:** Upload/download files, search by keywords/departments/years, version control for updates, reuse tracking, and reporting tools for faculty oversight.
- **Technical Stack Suggestion:** Frontend (HTML/CSS/JavaScript with React.js), Backend (Node.js or Python Flask/Django), Database (PostgreSQL or MongoDB for metadata), Storage (separate device with API integration for file handling), Authentication (OAuth/JWT for secure access).
- **Security and Compliance:** Role-based access control (RBAC), encryption for sensitive data (e.g., medical/biological datasets), audit logs, and backups to prevent data loss.

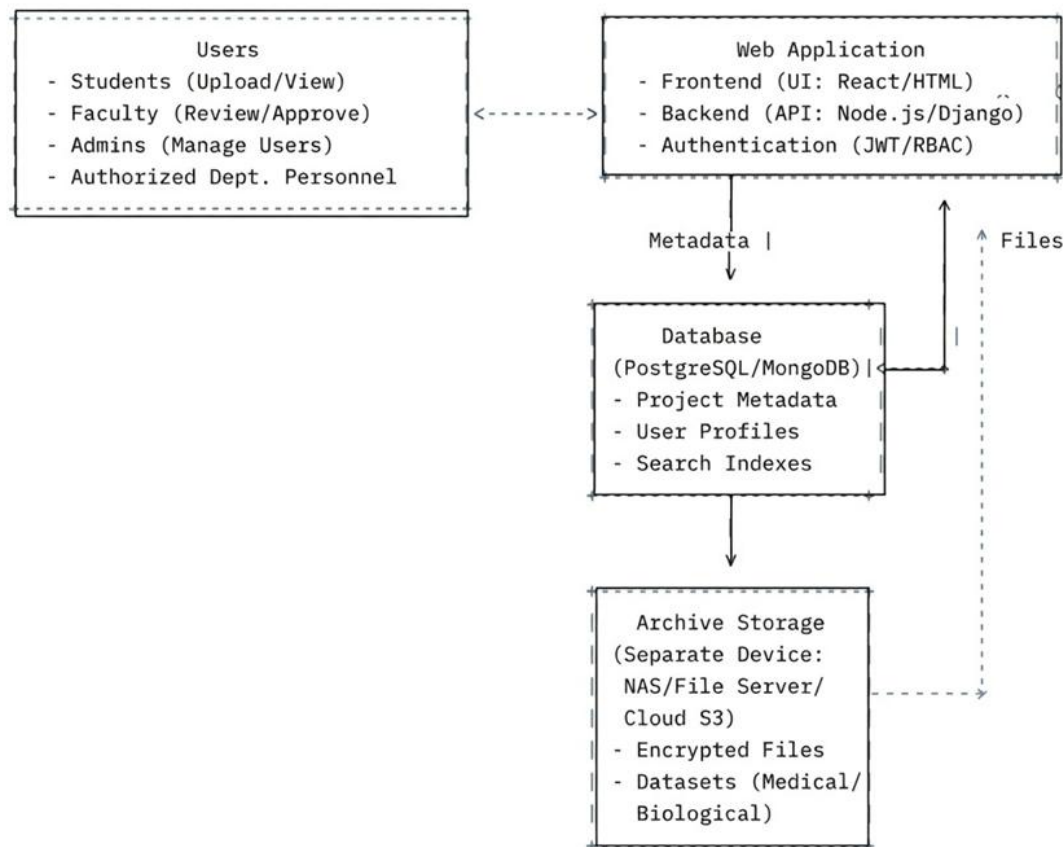


Figure 1: Overall System Architecture

Explanation of the above Diagram Components:

- Users interact via the web app (e.g., browsers on desktops/mobiles).

- Web Application handles requests, processes logic, and interfaces with the database for quick metadata access.
- Database stores structured info for fast searching/retrieval.
- Archive Storage is isolated for security and scalability—files are uploaded/downloaded via secure APIs (e.g., HTTPS with access tokens). Data flows bidirectionally: uploads go to storage, downloads/pulls come from it.
- Arrows indicate data flow; the separation ensures the web app isn't overloaded with large files.

3.2 Development Methodology

For building SPRAMS, an Agile methodology will be used due to the iterative nature of academic projects and the need for feedback from stakeholders (students, faculty, IT staff).

Key phases:

- **Planning and Requirements Gathering:** Collect input from faculty and students on features, data types, and ethical considerations.
- **Design:** Create wireframes, database schema, and architecture diagrams.
- **Development:** Build in sprints (e.g., 2-4 weeks), starting with core features like authentication and upload.
- **Testing:** Unit, integration, and user acceptance testing, focusing on security for data archives.
- **Deployment and Maintenance:** Host on a university server or cloud, with ongoing updates based on usage feedback.
- **Evaluation:** Monitor reuse metrics and gather post-implementation feedback.

3.3 Functional Requirements

- **User Management:** Allow users to register and log in with role-based access for Admin, Lecturer, and Student.
- **Project Management:** Enable Admin to add, edit, delete, and organize student project records by academic year, department, and group.

- Research Archive Management: Store and manage research information, including titles, descriptions, and related data.
- File Upload and Viewing: Allow students and Admin to upload and securely store project materials like proposals, slides, documents, audio, and video files.
- Project and Research Browsing: Allow users to browse and view archived projects and research by academic year or department.
- Search Functionality: Provide search and filtering to quickly find projects, research, or groups.
- Role-Based Access Control: Restrict add/edit/delete functions to Admin; allow view access for Lecturers and Students.
- Feedback Management: Allow lecturers to view project details and provide feedback.
- Notification System: Notify users of new projects, research data, or updates.
- Centralized Data Storage: Maintain all project, research, and files in a secure, organized database.
- Project Status Identification: Allow users to identify project status (completed, ongoing, unfinished) for reviewing past work and generating new ideas.

3.4 Non-functional Requirements

- Performance: Support multiple simultaneous users with smooth response times.
- Usability: Provide a simple, intuitive, responsive interface for desktops, tablets, and mobiles.
- Reliability: Ensure stable operation with minimal downtime and consistent access.
- Security: Implement secure authentication, role-based access, and protected data storage.
- Maintainability: Use modular design with clear documentation for updates.
- Portability: Compatible with major web browsers and devices.
- Data Integrity: Maintain accurate, consistent storage of information.
- Scalability: Handle growing numbers of projects, records, and users without performance loss.

3.5 Data Requirements

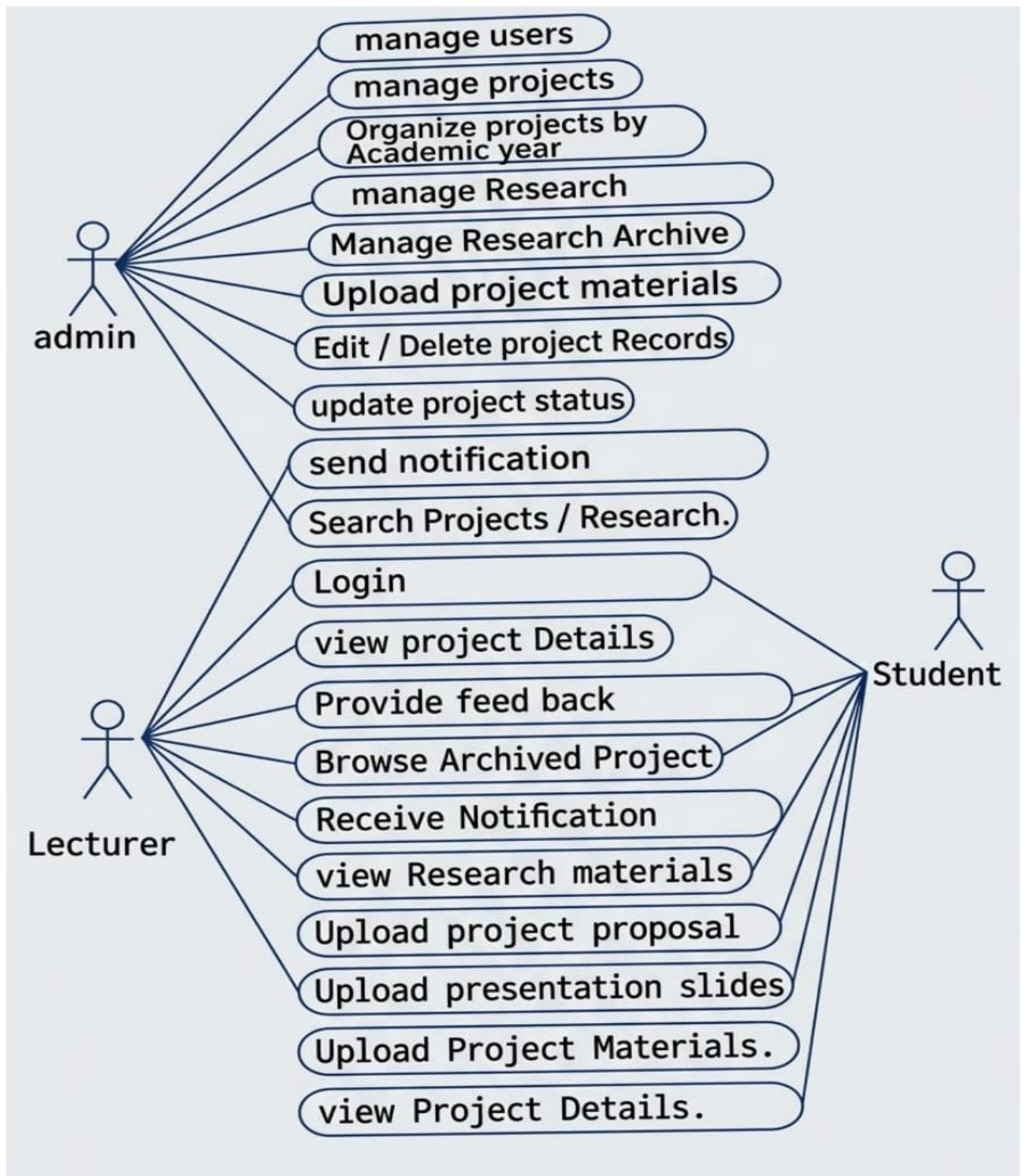
- Student [StudentID (PK), RegistrationNumber, Name, Email, Phone, DegreeProgram, Department, Batch, YearOfStudy, Status (Active/Completed)]

- Supervisor [SupervisorID (PK), Name, Email, Designation, Department, Specialization, ContactNumber]
- Project [ProjectID (PK), Title, Abstract, ProjectType (Group/Individual/Research), DegreeProgram, AcademicYear, Semester, Status (Proposed/Approved/Ongoing/Completed), SubmissionDate, CompletionDate]
- ProjectGroup (For Group Projects) [GroupID (PK), ProjectID (FK), GroupName, CreatedDate]
- GroupMember [GroupID (FK), StudentID (FK), Role (Leader/Member)] (Composite PK: GroupID + StudentID)
- Supervision [ProjectID (FK), SupervisorID (FK), Role (Main Supervisor/Co-Supervisor/Examiner)] (Composite PK: ProjectID + SupervisorID)
- Proposal [ProposalID (PK), ProjectID (FK), SubmittedDate, ApprovalStatus, Remarks, ApprovedBy, ApprovalDate]
- Submission [SubmissionID (PK), ProjectID (FK), SubmissionType (Proposal/Progress Report/Final Report), FilePath, SubmittedBy, SubmittedDate, VersionNumber]
- Evaluation [EvaluationID (PK), ProjectID (FK), EvaluatorID (FK → Supervisor), Marks, Feedback, EvaluationDate]
- ResearchPublication (Optional) [PublicationID (PK), ProjectID (FK), Title, JournalOrConference, DOI, PublicationYear, Indexed (Yes/No)]
- UserAccount (Access Control) [UserID (PK), Username, PasswordHash, Role (Student/Supervisor/Admin), LinkedID (StudentID or SupervisorID), LastLogin]
- Archive [ArchiveID (PK), ProjectID (FK), ArchivedDate, StorageLocation, AccessLevel (Public/Restricted/Internal)]

Key Relationships Summary

- One Project → Many Students (via GroupMember)
- One Project → Many Supervisors
- One Project → Many Submissions
- One Project → Many Evaluations
- One Project → Optional Publications
- One Project → One Archive Record

3.6 Use case diagram



3.3 Tools and Technologies

		Description
Layer / Purpose	Technology / Tool	
Frontend	React.js + Tailwind CSS	Provides a responsive and user-friendly web interface for students and lecturers to access the archive system across desktop and mobile devices.
Backend	Node.js	Handles server-side logic, user authentication, and processing of project and research data.
Database	MongoDB / MySQL	Stores student project records, research files, user information, and system data securely.
Development Environment	XAMPP / VS Code	Used for local server setup, coding, testing, and system development.
File Storage	Local host	Manages uploading and secure storage of documents, images, and multimedia research materials.
Security	Session Management + Role-Based Access	Ensures secure login, access control for students and lecturers, and protection of academic data.

4 Expected Outcome

The development and implementation of SPRAMS at the Faculty of Applied Science, University of Vavuniya, are expected to yield several tangible and intangible outcomes.

Improved Data Management and Organization

- **Centralized Repository:** All student projects, research outputs, datasets (e.g., medical or biological data with ethical approvals), documents, audio/video files, and other resources will be stored in a single, structured digital archive.
- **Efficient Tracking:** Project statuses (completed, ongoing, unfinished) will be identifiable, allowing for better oversight of academic activities across departments like Computer Science, Information Technology, and Environmental Science.

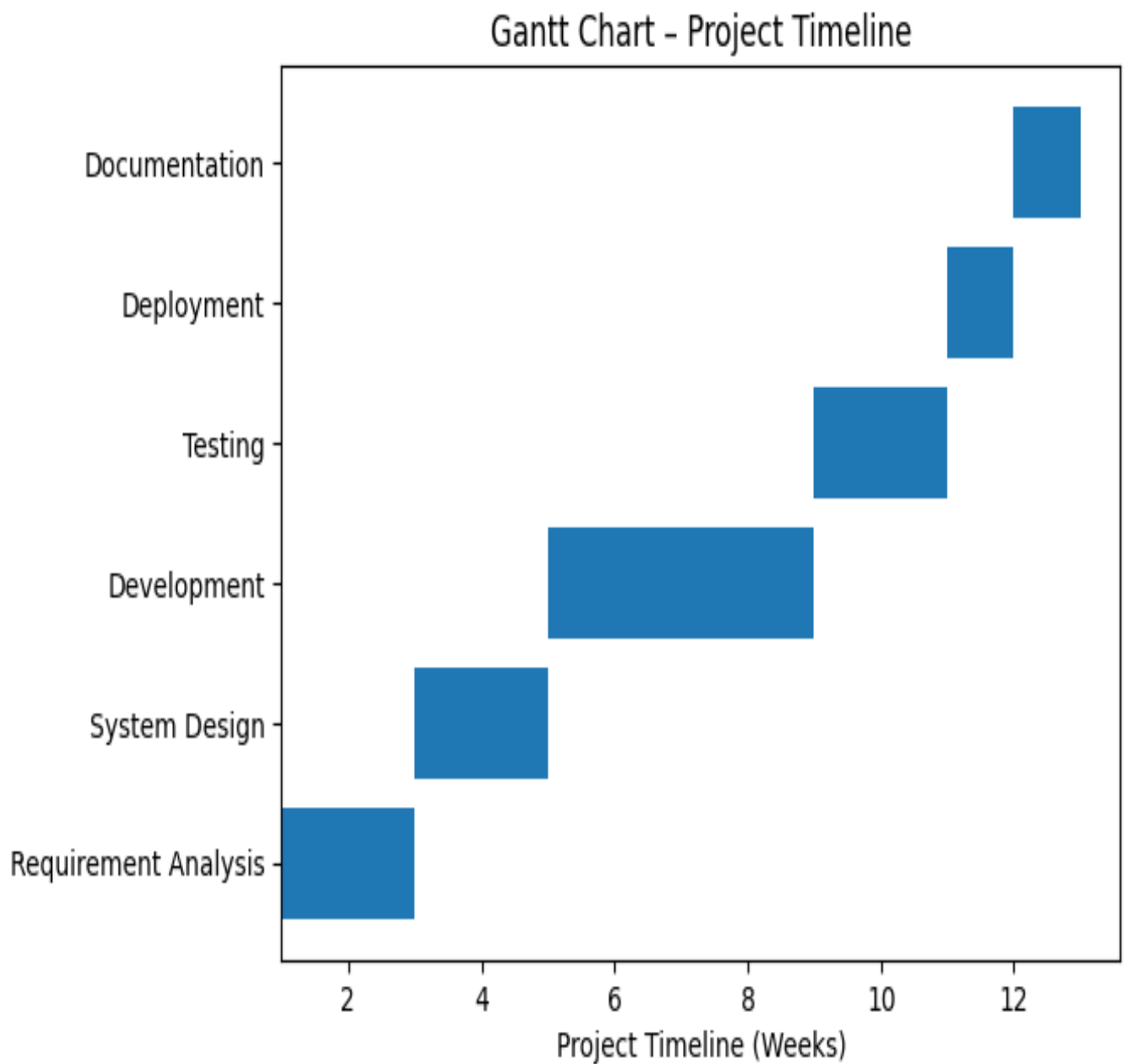
Operational Efficiency and User Engagement

- **Streamlined Processes:** Manual record-keeping will be replaced by automated functions, such as notifications for new additions, feedback submission by lecturers, and role-based access control (RBAC) to ensure security and proper permissions.
- **Increased Adoption:** With a simple interface and features like search functionality and project browsing, user satisfaction and engagement should rise, leading to higher submission rates of project materials and better compliance with archival policies.

Security, Enhanced Accessibility and Knowledge Sharing

- **Secure Storage:** Sensitive data will be housed on a separate storage device (e.g., NAS or cloud S3) with encryption, ensuring ethical standards for datasets are met and unauthorized access is prevented.
- **Long-Term Preservation:** The system will support data integrity and scalability, allowing the archive to grow with increasing projects without performance degradation, preserving institutional knowledge for years.
- **Resource Availability:** Features like links for source codes and file viewing will provide comprehensive insights into project implementations, promoting collaborative learning and reducing redundancy in research efforts.

5 Timeline of the Project



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